

Series Solution of ODEs — Frobenius Method

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Analytic Function

Definition (Analytic Functions)

A function f is said to be analytic at x_0 if, in an open interval about x_0 , the function is the sum of a power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ that has a positive radius of convergence.

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L.$$

Radius of convergence $R = \frac{1}{L}$.

* $L = \infty$, $R = 0$
* $L = 0$, $R = \infty$ } $L = \infty$ is not a good case.
 $(L < \infty)$

Recognizing Analytic Functions

How to recognize Analytic Functions?

Fortunately, we have some easy ways to recognize analytic functions. For example, if f, g are analytic at x_0 , so are $c_1 f + c_2 g$, fg , and f/g (when $g(x_0) \neq 0$) as well as their derivatives and integrals f' and $\int f$, respectively.

$$h(x) = \frac{1}{x}$$

Regularity

Definition (Ordinary & Singular Points)

Consider the linear differential equation in the standard form

$$y'' + p(x)y' + q(x)y = 0.$$

A point x_0 is called an ordinary point if both p, q are analytic at x_0 . If x_0 is not an ordinary point, it is called a singular point of the equation.

Theorem

Theorem (Existence of Analytic Solutions)

Suppose x_0 is an ordinary point for equation

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0,$$

then it has two linearly independent analytic solutions of the form

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

Moreover, the radius of convergence of any power series solution of the form given above is at least as large as the distance from x_0 to the nearest singular point (real or complex-valued).

Classification of Singular Points

Definition (Regular & Irregular Singular Point)

- If $p(x)(x - x_0)$, $q(x)(x - x_0)^2$ are analytic, then the solution can be solved via a generalized version of the series method; such x_0 is called **regular singular**.
- In all other cases, x_0 is called irregular singular, and there is no universally good way of solving the equation.

$$x^2 y'' + 2x y' + 6y = 0$$

$$x_0 = 0$$

$$\Rightarrow y'' + \frac{2}{x} y' + \frac{6}{x^2} y = 0 \quad (x \neq 0)$$

$$p(x) = \frac{2}{x}, \quad q(x) = \frac{6}{x^2} \quad \left(\begin{array}{l} p \text{ and } q \text{ both} \\ \text{are not analytic at } x_0 = 0 \end{array} \right)$$

$$P_1(x) = p(x)x = x \cdot \frac{2}{x} = 2, \quad \left\| \begin{array}{l} q_1(x)x^2 = \frac{6}{x^2} \cdot x^2 = 6 \\ q_2(x) = \frac{6}{x^2} \end{array} \right.$$

Series Solution Near a Singular Point

$$\underline{y'' + p(x)y' + q(x)y = 0.} \quad x = x_0 \text{ a singular point}$$

- We have already studied the Euler type equations

$$(x - x_0)^2 y'' + \alpha(x - x_0)y' + \beta y = 0$$

$$p(x) = \frac{\alpha}{x - x_0}$$

$$q(x) = \frac{\beta}{(x - x_0)^2}$$

or

$$y'' + \frac{\alpha}{x - x_0} y' + \frac{\beta}{(x - x_0)^2} y = 0$$

where α, β are constants was to prepare us for the study of power series solution to differential equations of the type

$$y'' + p(x)y' + q(x)y = 0$$

$$p(x) = \frac{\alpha}{x - x_0} +$$

when x_0 is not an ordinary point.

$$y'' + \underbrace{\left(\frac{\alpha}{x - x_0} + x\right)}_{r(x)} y' + \underbrace{\left(\frac{\beta}{(x - x_0)^2} + x^2\right)}_{s(x)} y = 0.$$

$$s(x) = \frac{\beta}{(x - x_0)^2} + x^2$$

Qn: Are $r(x)$ and $s(x)$ analytic at $x=x_0$?

$$\left. \begin{aligned} r(x) &= \boxed{\frac{\alpha}{x-x_0}} + \textcircled{x} \\ s(x) &= \boxed{\frac{\beta}{(x-x_0)^2}} + \textcircled{x^2} \end{aligned} \right\}$$

$$(x-x_0) r(x) = \alpha + \underbrace{x(x-x_0)}_{\text{analytic}}$$

$$(x-x_0)^2 s(x) = \beta + \underbrace{x^2(x-x_0)^2}_{\text{analytic}}$$

Introduction to the Method

- Suppose x_0 is a regular singular point of $(y'' + p(x)y' + q(x)y) = 0$. $\times (x-x_0)^2$
- Multiplying the equation by $(x - x_0)^2$, we get

$$(x - x_0)^2 y'' + (x - x_0) [(x - x_0)p(x)] y' + (x - x_0)^2 q(x)y = 0$$

or

$$(x - x_0)^2 y'' + (x - x_0)p_1(x)y' + q_1(x)y = 0$$

analytic at x_0

where

$$(x - x_0)p(x) = p_1(x) \text{ and } (x - x_0)^2 q(x) = q_1(x).$$

- The simplest type of equation that falls into this category is the second-order Cauchy-Euler equation

$$x^2 y'' + p_0 x y' + q_0 y = 0,$$

where p_0, q_0 are constants.

$$\begin{aligned} p_1(x) &= p_0 + p_1 x + p_2 x^2 + \dots \\ q_1(x) &= q_0 + q_1 x + q_2 x^2 + \dots \end{aligned}$$

$$x^2 y'' + a x y' + b y = 0$$

$$\Rightarrow y'' + \frac{a}{x} y' + \frac{b}{x^2} y = 0 \quad (x \neq 0)$$

$$p(x) = \frac{a}{x} \quad \& \quad q(x) = \frac{b}{x^2}$$

$$x p(x) = a = p_1(x) = \boxed{a} + 0 \cdot x + 0 \cdot x^2 + \dots$$

$$x^2 q(x) = x^2 \cdot \frac{b}{x^2} = b = \boxed{b} + 0 \cdot x + 0 \cdot x^2 + \dots$$

Frobenius Method

- Recall that the above equation has a solution of the form $y = x^r$, where r is the root of the indicial equation

$$r(r-1) + p_0 r + q_0 = 0.$$

- Since p_1 and q_1 are analytic at $x = x_0 = 0$ (say), we can write

and

$$\begin{cases} p_1(x) = c_0 + c_1 x + c_2 x^2 + \dots \\ q_1(x) = d_0 + d_1 x + d_2 x^2 + \dots \end{cases}$$

$\underbrace{\hspace{1.5cm}}_{p_0}$ $\underbrace{\hspace{1.5cm}}_{q_0}$

$x^r \sum_{n=0}^{\infty} a_n x^n$

- Then we can get from the ODE

$$x^2 y'' + x(c_0 + c_1 x + c_2 x^2 + \dots) y' + (d_0 + d_1 x + d_2 x^2 + \dots) y = 0.$$

- In a suitable interval ($|x| \ll 1$) this approximates the Cauchy-Euler equation.

Frobenius Method

- The Frobenius method is a modification of the power series method guided by the above observation. This method is effective at regular singular points. The basic idea is to look for solutions of the form

$$(x - x_0)^r \sum_{n=0}^{\infty} a_n (x - x_0)^n \quad (1)$$

where r is a root of the indicial equation

$$r(r - 1) + p_0 r + q_0 = 0.$$

- A series of the form (1) is called a Frobenius Series.

Indicial Equation:- Let x_0 be a regular singular point for the ODE

$$y'' + p(x)y' + q(x)y = 0,$$

For simplicity take $x_0 = 0$.

Then, the indicial equation is given by

$$r(r-1) + p_0 r + q_0 = 0.$$

where p_0 and q_0 are coming from the power series representation

① Theorem 1: Suppose that $x_0 = 0$ is a regular singular point of

$$y'' + p(x)y' + q(x)y = 0,$$

and r_1 and r_2 are the roots of its indicial equation with $r_1 > r_2$. If $r_1 - r_2$ is not a positive integer, then this differential equation has a solⁿ of the form

$$y_1 = x^{r_1} \sum_{n=0}^{\infty} a_n(r_1) x^n = x^{r_1} \sum_{n=0}^{\infty} a_n x^n.$$

and a second solⁿ of the form

$$y_2 = x^{r_2} \sum_{n=0}^{\infty} a_n(r_2) x^n = x^{r_2} \sum_{n=0}^{\infty} b_n x^n.$$

② Theorem 2: For the same ODE, if the indicial equation has roots $r_1 > r_2$, where $r_1 - r_2$ is a positive integer. Then, this differential eqnⁿ has a solⁿ of the form

$$y_1(x) = x^{r_1} \sum_{n=0}^{\infty} a_n(r_1) x^n$$

and there is a constant a so that it has a second solⁿ of the form:

$$y_2 = \underbrace{a y_1}_{\text{constant}} \underbrace{\ln|x|} + x^{r_2} \sum_{n=1}^{\infty} a_n(r_2) x^n.$$

Theorem 3:- Next, suppose that for the equation under the discussion, has a root r_1 of multiplicity 2 for the indicial equation. Then, the differential equation has a solⁿ of the form

$$y_1(x) = x^{r_1} \sum_{n=0}^{\infty} a_n x^n,$$

and a second solⁿ of the form

$$y_2 = y_1 \ln x + x^{r_1} \sum_{n=0}^{\infty} b_n x^n. //$$

Example

Example

Find the general solution to around $x_0 = 0$

$$2x^2 y'' + xy' - (1+x)y = 0, \quad x > 0.$$

$$\Rightarrow y'' + \frac{x}{2x^2} y' - \frac{(1+x)}{2x^2} y = 0 \quad (x > 0)$$

$$\Rightarrow y'' + \frac{1}{2x} y' - \left(\frac{1+x}{2x^2}\right) y = 0$$

$$p(x) = \frac{1}{2x}, \quad \text{and} \quad q(x) = -\left(\frac{1+x}{2x^2}\right) \quad (\text{by comparison})$$

$p(x)$ and $q(x)$ are not analytic at $x=0$.

$$\text{But } \underline{x p(x)} = \frac{1}{2} \quad \text{and} \quad x^2 q(x) = -\left(\frac{1+x}{2}\right).$$

$$x^2 p(x) = \left(\frac{1}{2}\right) \Rightarrow p_0 = 1/2, p_1 = 0, p_2 = 0, \dots$$

$$x^2 q(x) = \left(-\frac{1}{2}\right) \cdot \frac{x}{2} = a_0 + a_1 x + a_2 x^2 \quad \parallel \quad a_0 = -\frac{1}{2}, a_1 = -\frac{1}{2}$$

$$a_2 = a_3 = \dots = 0$$

The indicial equation is

$$r(r-1) + \frac{1}{2}r - \frac{1}{2} = 0 \quad (r(r-1) + p_0 r + a_0 = 0)$$

$$\Rightarrow r^2 - r + \frac{1}{2}r - \frac{1}{2} = 0$$

$$\Rightarrow r^2 - \frac{1}{2}r - \frac{1}{2} = 0 \quad \Rightarrow r = 1, -\frac{1}{2}$$

Theorem 1 will be applicable: $r_1 = 1$ and $r_2 = -\frac{1}{2}$

and also $r_1 - r_2 = 1 + \frac{1}{2}$ is not a positive integer.

The first solution is

$$y_1(x) = x \sum_{n=0}^{\infty} a_n(1) x^n$$

The second solⁿ would be

$$y_2(x) = x^{-1/2} \sum_{n=0}^{\infty} a_n (-1/2) x^n$$

$$y_1(x) = x \sum_{n=0}^{\infty} a_n (1) x^n = \sum_{n=0}^{\infty} a_n (1) x^{n+1}$$

$$y_1', y_1''$$

$$y_2(x) = \sum_{n=0}^{\infty} a_n (-1/2) x^{n-1/2} = a_0 x^{-1/2} + a_1 x^{1/2} + a_2 x^{3/2} + \dots$$

$$y_2' = \sum_{n=0}^{\infty} (n-1/2) a_n x^{n-3/2}$$

$$= -\frac{1}{2} a_0 x^{-1/2-1} + \frac{1}{2} a_1 x^{1/2-1} + a_2 \frac{3}{2} x^{3/2-1} + \dots$$

$$y_2'' = \sum_{n=0}^{\infty} (n-1/2) a_n (n-3/2) x^{n-5/2}$$

$$a_n =$$

Bessel's Equation:- Bessel f^n are solⁿ to the eqnⁿ

$$x^2 y'' + xy' + (x^2 - \nu^2) y = 0,$$

where ν is a parameter.

Consider $\boxed{\nu=1}$, then we get

$$x^2 y'' + xy' + (x^2 - 1) y = 0.$$

$$\Rightarrow y'' + \frac{x}{x^2} y' + \frac{(x^2 - 1)}{x^2} y = 0$$

$$= y'' + \frac{1}{x} y' + \left(1 - \frac{1}{x^2}\right) y = 0$$

$$x^2 q(x) = x^2 - 1; \quad q_0 = -1 \quad \left| \quad x \quad p(x) = 1 = p_0 \right|$$

$$p(x) = \frac{1}{x}$$

$$q(x) = 1 - \frac{1}{x^2}$$

$x=0$ is singular
but regular

$$r(r-1) + r - 1 = 0$$

$$\Rightarrow r^2 - r + r - 1 = 0.$$

$$\Rightarrow r^2 = 1.$$

● Example: Determine the series solⁿ to the following Bessel equation: the Bessel equation of order 1,

$$x^2 y'' + xy' + (x^2 - 1)y = 0.$$

Solⁿ: Here the indicial equation's roots are $r_1 = 1$ and

$$r_2 = -1.$$

$$r^2 = 1, \quad r_1 = 1, \quad r_2 = -1$$

$$r_1 - r_2 = 2.$$

Theorem 2 will be applicable:

First solⁿ $y_1(x) = x \sum_{n=0}^{\infty} a_n \underbrace{x^n}_{\checkmark} = \sum_{n=0}^{\infty} a_n x^{n+1}.$

$y_2(x) \Rightarrow$ difficult.

$$J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+1)!} \left(\frac{x}{2}\right)^{2k+1} = y_1(x)$$

Second solⁿ ———

Thank You

Questions?